

TEST 2

Course: Applied Statistics for Engineering and Scientists 1 **Course ID:** STAT451

Duration: 90 minutes

Materials: One A4 formula sheet allowed

Student name: **Student ID:** **No.:**

Note:

- All calculations must be rounded to **four** decimal places.
- Standard normal $\mathcal{N}(0, 1)$ and Student t distribution tables are provided.

Question 1. (2.0 *points*) The diameter of stainless steel rods manufactured on an extruder is normally distributed with a mean of $\mu = 0.503$ inch and a standard deviation of $\sigma = 0.004$ inch. Rods are classified as acceptable only if their diameter falls in the interval $[0.495, 0.510]$ inch.

- (a) Compute the probability that a randomly selected rod is *not* acceptable (i.e., is defective).
- (b) Among a random sample of $n = 120$ rods, let X denote the number of defective rods. Using the normal approximation to the binomial distribution (with continuity correction), find $P(X \geq 10)$.
- (c) A random sample of $n = 25$ rods is drawn; let $\bar{X}$ be the sample mean diameter. Determine the probability that $\bar{X}$ differs from the process mean by less than 0.001 inch.

Question 2. (2.0 *points*) A random sample of $n = 15$ specimens of a particular type of aluminum alloy yielded the following yield strengths (in ksi):

33.8, 35.6, 34.2, 36.1, 33.4, 35.0, 34.7, 36.3,
33.9, 35.2, 34.5, 35.9, 33.6, 34.8, 36.0

Assume the yield strengths are approximately normally distributed.

- (a) Compute the sample mean $\bar{x}$ and sample standard deviation s . Then construct a 98% confidence interval for the true mean yield strength μ .
- (b) In a quality audit, 186 out of 250 randomly inspected aluminum castings passed all surface-defect tests. Construct a 95% confidence interval for the true proportion p of castings that pass.
- (c) Using the preliminary estimate from part (b), how many castings should be inspected to estimate p to within 0.03 with 95% confidence?

Question 3. (2.0 *points*) The specification sheet for a certain chemical process asserts that its mean yield is exactly 500 grams per batch. A process engineer suspects that recent production has drifted from this target. A random sample of $n = 16$ batches produced a sample mean yield of $\bar{x} = 493.4$ grams with a sample standard deviation of $s = 11.2$ grams. Yields are approximately normally distributed.

- (a) At the $\alpha = 0.05$ significance level, test the hypothesis $H_0 : \mu = 500$ against $H_1 : \mu \neq 500$. State the rejection region, the test statistic, and your conclusion.
- (b) In a separate study, the engineer records that 58 out of 400 randomly selected batches failed to meet the yield target. The company had previously claimed the true failure rate is at most 12%. Test this claim at the $\alpha = 0.01$ significance level using the rejection-region approach.
- (c) Compute the p -value for part (b) and interpret its meaning.

Question 4. (1.5 *points*) Two machines are used to fill plastic bottles with a net volume of 16.0 ounces. The fill volumes are assumed to be normal. To compare the two machines, quality engineers collect random samples with the following results:

	Machine 1	Machine 2
Sample size	$n_1 = 10$	$n_2 = 10$
Sample mean	$\bar{x}_1 = 16.015$	$\bar{x}_2 = 16.005$
Sample standard deviation	$s_1 = 0.030$	$s_2 = 0.025$

- (a) Assuming the two population variances are *equal*, test at $\alpha = 0.05$ whether there is any difference in the mean fill volumes.
- (b) Construct a 95% confidence interval for the difference in mean fill volumes, $\mu_1 - \mu_2$.

Question 5. (2.5 *points*) A researcher investigates the relationship between the hardness (y , Rockwell B) of a copper alloy and the percentage of tin (x , %) added to the alloy. Data for $n = 8$ specimens were recorded:

x	0.5	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	41.2	45.0	48.7	52.4	55.8	59.1	62.3	65.4

- (a) Compute the sample correlation coefficient r between tin percentage and hardness, and interpret it.
- (b) Determine the least-squares regression line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$, and interpret both estimated coefficients.
- (c) Determine the coefficient of determination R^2 and interpret it in the context of this study.
- (d) Predict the hardness when the tin percentage is $x = 2.75\%$ and when $x = 6.0\%$. For each prediction, state whether the estimate is reliable and justify your answer.

— END OF TEST 2 —